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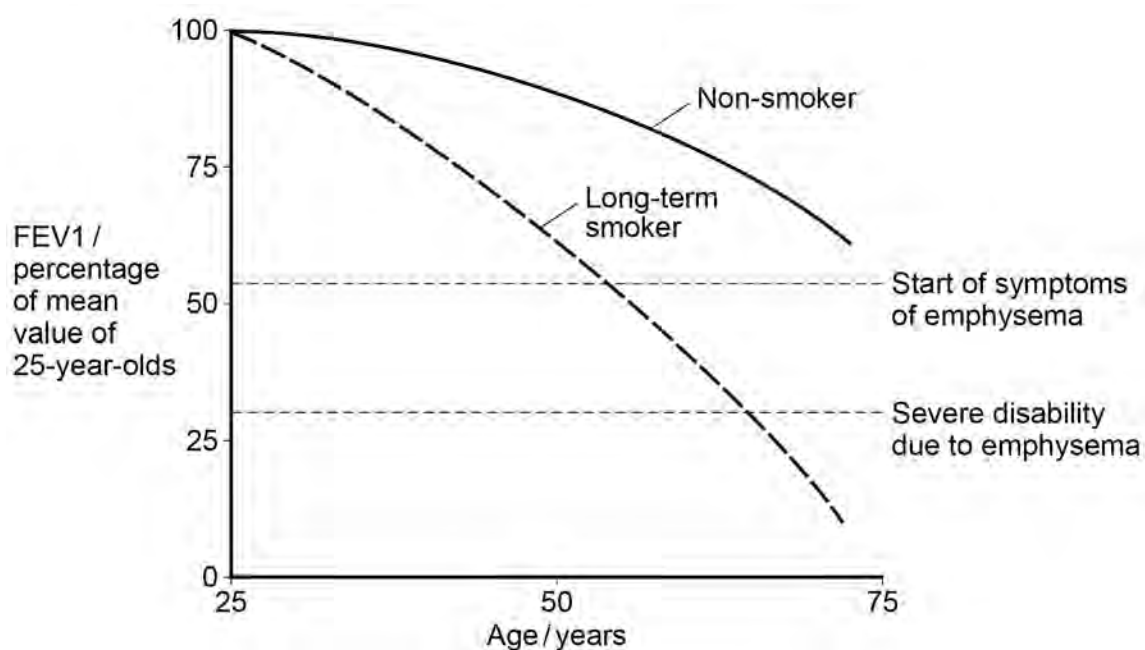
Forced expiration volume (FEV1) is the volume of air a person can breathe out in 1 second.

Emphysema is a lung disease which results in a reduction in FEV1. Emphysema is mainly caused by long-term cigarette smoking.

Scientists investigated the effects of ageing and long-term cigarette smoking on FEV1 and on the development of emphysema.

Figure 7 shows their results.

Figure 7



0 6 . 1

Scientists determined the mean FEV1 value of 25-year-olds in the population.

Suggest **two** precautions that should have been taken to ensure that this mean FEV1 value was reliable.

[2 marks]

1 _____

2 _____



0 6 . 2

Explain the importance of determining a mean FEV1 value of 25-year-olds in this investigation.

[2 marks]

0 6 . 3

The mean FEV1 value of non-smokers decreases after the age of 30.

Use your knowledge of ventilation to suggest why.

[1 mark]

0 6 . 4

One of the severe disabilities that results from emphysema is that walking upstairs becomes difficult.

Explain how a low FEV1 value could cause this disability.

[3 marks]

